

Flow Matching for Molecular Generation

Thesis Report 2

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The implementation lives at github.com/Nico-Conti/flow-matching-molecules.

1 Quick review of the ZINC pipeline

The ZINC featurization was changed slightly after reviewing the DeFoG setup (github.com/manuelmlmadeira/DeFoG).

Previously we kept the non-neutral groups N^+ and O^- as standalone atom types. The node vocabulary was therefore charge-aware, with 11 tokens, using a separate one-hot encoding for these charged species:

$$\{\text{C}, \text{N}, \text{O}, \text{F}, \text{P}, \text{S}, \text{Cl}, \text{Br}, \text{I}, \text{N}^+, \text{O}^-\}.$$

Following the DeFoG implementation, the charge-aware tokens are dropped and the model works over an element-only, neutral vocabulary of 9 types,

$$\{\text{C}, \text{N}, \text{O}, \text{F}, \text{P}, \text{S}, \text{Cl}, \text{Br}, \text{I}\}.$$

Charged molecules are not skipped, their bond structure is kept, but every atom is represented neutrally, and the charge is recovered at decode time by a partial-charge reconstruction.

The decoder builds the molecule one bond at a time and checks valency after each bond. When an N, O, or S atom ends up over its neutral valence by exactly one (N:3, O:2, S:2), that excess is read as a positive formal charge and the atom is set to +1. This recovers cations such as N^+ purely from the bonding pattern, with no dedicated vocabulary entry. Anions such as O^- are not recovered by this rule and decode as the corresponding neutral acid; doubly-charged or otherwise charge-unbalanced atoms are likewise not represented.

To quantify how many SMILES are reconstructable after being encoded without charge: every SMILES is reduced to its charge-neutral graph and then rebuilt by the partial-charge decoder, and a molecule is counted as usable only if its original charged form is recovered exactly. 247,449 of 249,455 molecules (99.20%) remain usable.

2 DeFoG implementation

The new method reuses the backbone: the graph transformer carrying node features $\mathbf{X}$, a dense edge tensor $\mathbf{E}$, and a global vector y .

The discrete objective. DeFoG keeps the state on the discrete graph simplex and treats generation as a continuous-time Markov jump process (CTMC): the network predicts the clean-graph distribution and sampling moves between discrete states along a learned rate matrix.

Sampling. Generation integrates the rate matrix

$$R = R^* + \eta p_t.$$

Following DeFoG, a sampling-time distortion scheduler was implemented (`identity` or `polydec`) that re-allocates integration steps over $[0, 1]$.

The prior at t_0 is taken to be uniform, with no experimentation using the dataset marginal or other distributions, and no train-time distortion has been applied.

3 Structural features

The ZINC experiments used extra structural input features, such as:

- RRWP (relative random-walk positional encodings);
- graph cycle counts (rings of length 3–6), fed through the y stream.

4 Results

QM9

The DeFoG run uses `polydec` distortion with $\eta = 0$; no RRWP or graph cycle counts were used during training.

Model	Validity	Uniqueness	FCD↓
<i>Paper reference</i>			
CatFlow [1]	99.8	100.0	0.44
Flow matching [1]	94.10	98.20	5.16
DeFoG (50 steps) [2]	98.9	96.2	0.26
DeFoG (500 steps) [2]	99.3	96.3	0.12
<i>My results</i>			
Flow matching (100 steps)	95.81	96.11	1.05
DeFoG (50 steps)	97.75	96.75	0.15
DeFoG (500 steps)	98.38	96.30	0.13

[1] from Table 2, Eijkelboom et al.; [2] from Table 8, Qin et al.

ZINC-250k

Training used 20 RRWP steps and graph cycle counts; the best sampling η was found to be 300.

Model	Validity	Uniqueness	FCD↓
<i>Paper reference</i>			
CatFlow [1]	99.21	100.00	13.21
Flow matching [1]	94.01	96.68	18.76
DeFoG (50 steps) [2]	96.65	99.99	2.12
DeFoG (500 steps) [2]	99.22	99.99	1.43
<i>My results</i>			
Flow matching (200 steps)	91.88	99.99	16.93
DeFoG (50 steps)	93.41	99.96	7.76
DeFoG (500 steps)	98.70	99.66	3.38

[1] from Table 2, Eijkelboom et al.; [2] from Table 11, Qin et al.